

GROWRIDER · DESIGN PRECIS

GrowRider design precis

Open Hardware Portfolio · amishchadha.com

DOCUMENT CONTROL

Document ID	GRR-PRC-001	Version	0.2
Status	Draft	Date	2026-09-24
Author	Amish Chadha	License	CERN-OHL-S-2.0
Project TRL	TRL 2 Technology concept formulated		
Repository	github.com/BoujeeEnjinia1701/growrider		
Source	docs/02-concept.md @ uncommitted		

REVISION HISTORY

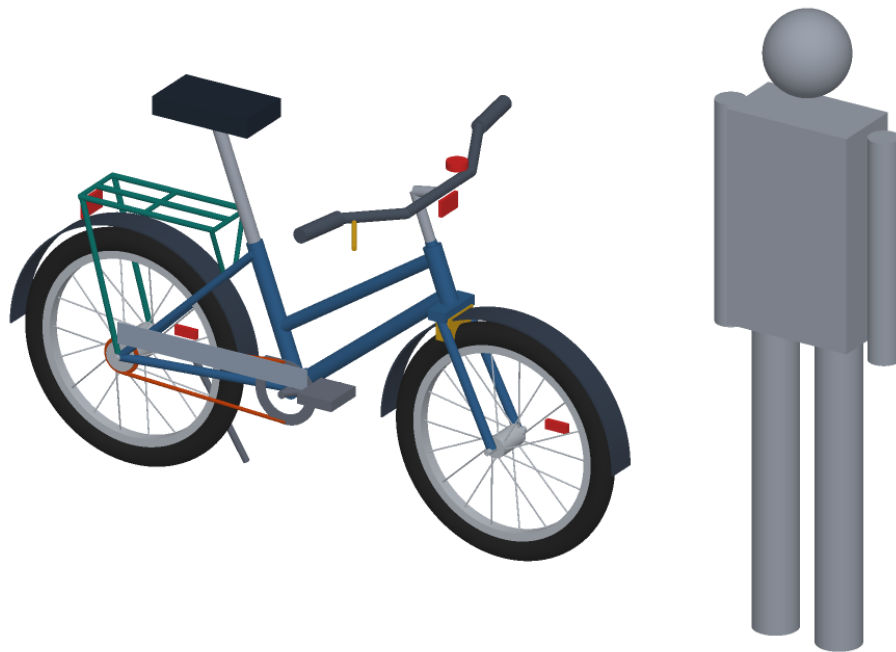
Version	Date	Author	Change
0.2	2026-09-24	Amish Chadha	Populate to TRL 2 (architecture, first-order numbers, design choices, safety, media)
0.1	2026-09-24	Amish Chadha	Initial scaffold

GrowRider design precis

GrowRider is a step-through steel children's bicycle on 20 in wheels whose seat post and stem telescope, so one bike fits riders from about 1.10 to 1.65 m (ages about 6 to 14) and can be handed down between siblings. It uses a coaster brake plus a front rim brake, puncture-proof tires, a single-speed chain drive and a small rack rated 10 kg, with wear parts shared with the region's adult roadsters. First-order numbers show the fit range, brakes and cost are achievable; mass (about 15.5 kg against a 13 kg target) is the main problem to solve.

GrowRider

CONCEPT, NOT FOR FABRICATION



Grey: Child, 1.35 m for scale

Figure 1. GrowRider massing model with the seat post and stem extended toward the top of their range, beside a 1.35 m child (about 10 years old) for scale.

■ How it works

- Fit the rider.** A parent, teacher or mechanic sets saddle height with a two-stage telescoping seat post (an inner sleeve inside the seat tube, then a standard seat post inside the sleeve) and sets handlebar height with a long quill stem that slides in the fork steerer. Minimum insertion marks on each sliding part show the limit. Both clamps use 13 mm bolts.
- Ride to school.** The rider pedals a single-speed chain drive (32T chainring, 18T sprocket, 140 mm cranks) at about 65 rpm for 11 km/h. Solid or airless tires cannot go flat on thorns.

- 3. **Stop.** Pedaling backward engages the coaster brake in the rear hub. A front rim brake with a short-reach lever gives a second, independent brake that still works if the chain comes off.
- 4. **Carry a school bag.** A small rear rack, rated and marked 10 kg, carries a bag or books. It is sized for a bag, not a passenger or a water container.
- 5. **Grow and hand down.** As the child grows, the post and stem are raised. When the child moves to an adult roadster, both are lowered and the bike passes to a younger sibling. Worn parts are replaced from the roadster parts already sold in the local market.

GrowRider: growth and hand-down cycle (estimates)

CONCEPT, NOT FOR FABRICATION

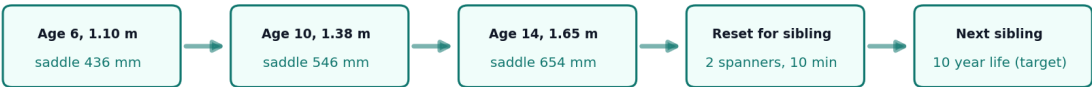


Figure 2. Growth and hand-down cycle. Saddle heights are estimates from the fit rules in the first-order numbers; the 10-year life is a target.

■ Main components

Item numbers match the exploded view (Figure 3) and bom/bom.csv .

#	Component	Proposed choice	Notes
1	Frame	Step-through steel frame with twin down tubes; 31.8 mm outer seat tube	Low stepping point for small riders and for school skirts; sleeves for the telescoping parts. Proposed, awaiting Amish
2	Fork	20 in steel fork, 1 in threaded steerer	Standard 1 in headset as on roadsters
3	Telescoping seat post	28.6 mm inner sleeve plus 25.4 mm seat post	218 mm of travel in two stages of about 110 mm
4	Saddle	Child size, light color cover	Light color stays cooler in sun
5	Telescoping stem	Long 22.2 mm quill, 120 mm travel, two-position head (0 or 60 mm forward)	Adjusts bar height and reach
6	Handlebar and grips	22.2 mm swept-back bar, 520 mm wide	Upright position for visibility
7	Front wheel	20 in (ISO 406), 28 spokes, nutted axle	15 mm axle nuts as on roadsters
8	Rear wheel	20 in (ISO 406) with coaster brake hub	Hub type widely used on regional roadsters (to confirm)
9	Tires (2)	20 x 1.95 in solid or airless	Proposed, awaiting Amish
10	Crankset	140 mm cranks, 32T ring, 9/16 in pedal threads	Adult pedals fit
11	Chain and sprocket	1/2 x 1/8 in chain, 18T sprocket	Same chain as roadsters
12	Front brake	Side-pull caliper, short-reach adjustable lever	Second, independent brake
13	Rear rack	Steel, about 300 x 120 mm, 10 kg marked	No footrests or passenger seat
14	Fenders	Pair, full coverage	Rainy-season mud and school uniforms
15	Chainguard	Full plate over the upper chain run	Keeps clothing out of the chain
16	Kickstand	Rear-mounted	Parking at school
17	Reflectors and bell	Front white, rear red, spoke and pedal reflectors, bell	Road visibility

GrowRider: exploded view

CONCEPT, NOT FOR FABRICATION

- 1 Step-through steel frame
- 2 Fork
- 3 Telescoping seat post (extended)
- 4 Saddle
- 5 Telescoping stem
- 6 Handlebar and grips
- 7 Front wheel, 20 in
- 8 Rear wheel, coaster brake hub
- 9 Puncture-proof tires (2)
- 10 Crankset, 32T, 140 mm
- 11 Chain and 18T sprocket
- 12 Front rim brake and lever
- 13 Rear rack, 10 kg rated
- 14 Fenders
- 15 Chainguard
- 16 Kickstand
- 17 Reflectors and bell

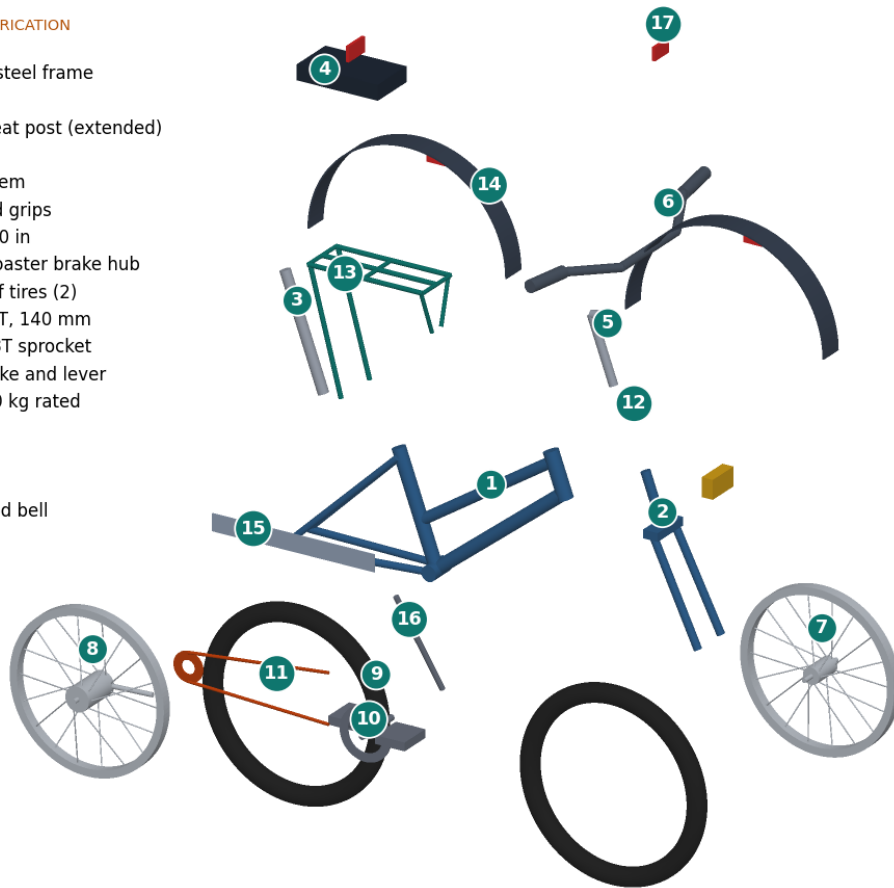


Figure 3. Exploded view with item numbers matching the BOM. The bottom bracket, headset, hardware and helmet (BOM lines 18 to 20) are not modeled.

■ First-order numbers

All values are estimates for concept review and will be checked at TRL 3.

Fit. Assumptions: inseam is about 0.45 times standing height; saddle height (bottom bracket center to saddle top) is about 0.88 times inseam; seat tube angle 70°.

Rider height	Inseam (0.45 x height)	Saddle height (0.88 x inseam)
1.10 m (about age 6)	495 mm	436 mm
1.38 m (about age 10)	621 mm	546 mm
1.65 m (about age 14)	743 mm	654 mm

Table 1. Saddle height across the fit range (estimates).

Quantity	Estimate	Basis	Requirement
Saddle height range	436 to 654 mm (218 mm)	Table 1	R2, met by two stages of about 110 mm with 100 mm insertion each
Saddle rise and setback over the range	205 mm up, 75 mm back	218 mm along a 70° seat tube	
Handlebar height range	about 113 mm	120 mm of quill travel along a 70° steering axis	R3 met
Reach adjustment	about 124 mm	Saddle setback 75 mm, less 41 mm as the bar rises back along the steering axis, plus 60 mm stem head and 30 mm saddle rail	R3 met; needs about 110 mm if reach is 0.2 x height (assumption)
Standover at the stepping point	about 460 mm	Low loop tube, 20 in wheels, 260 mm bottom bracket height	R1 met, thin margin; about 510 mm with 24 in wheels (not met)
Minimum saddle top above ground	about 670 mm	260 mm bottom bracket plus 436 mm at 70°	Small riders stand over the frame to stop, as on any bicycle
Pedal clearance	about 110 mm	260 mm bottom bracket, 140 mm cranks	Adequate for ruts; check on field roads
Gear development	2.8 m per crank turn	32/18 on a 1.57 m rolling circumference	66 rpm at 11 km/h
Power on the level at 11 km/h	about 35 W	45 kg rider and bike, rolling coefficient 0.02 (solid tire on gravel), small air drag	Sustainable for a child
Power on a 5 % climb at 7 km/h	about 60 W	Same, plus gravity	Short climbs only; steep hills walked
Mass	about 15.5 kg	Frame 3.5, wheels 2.6, solid tires 2.0, drive 1.7, rack 0.8, other parts 4.9 kg	R5 (13 kg) not met
Rear coaster brake alone, dry	about 0.27 g	Rear-wheel skid limit: $\mu (L - x) / (L + \mu h)$, $\mu = 0.7$, wheelbase $L = 0.86$ m, center of mass $x = 0.33$ m ahead of rear axle and $h = 0.70$ m high	R7 met
Rear coaster brake alone, loose gravel	about 0.19 g	Same with $\mu = 0.4$; 4.7 m stop from 15 km/h	R7 marginal; the case for a front brake
Front brake pitch-over limit	about 0.76 g	$(L - x) / h$	Grip limits both brakes to about μ , so 0.4 g on gravel and above 0.35 g on dry ground if a child can squeeze hard enough
Rack	10 kg	Rating chosen for child safety; school bag 3 to 5 kg	R6
Prototype parts cost	about \$243 for the bike, \$255 with helmet	Indicative prices, see bom/bom.csv	R11 met for the bike; \$5 over with the helmet

Commute time. Assumptions: a 5 km trip each way; walking at 4 to 5 km/h; cycling at 10 to 12 km/h on dirt roads.

Mode	Speed	One way	Round trip
Walking	4 to 5 km/h	60 to 75 min	2.0 to 2.5 h
GrowRider	10 to 12 km/h	25 to 30 min	50 to 60 min
Time returned to the child		30 to 50 min	60 to 100 min per day, about 250 h per 190-day school year at the midpoint

Table 2. School trip time, walking and cycling (estimates).

■ Key design choices

- **Wheel size: 20 in (ISO 406) or 24 in (ISO 507).** 20 in keeps standover at about 460 mm so a 1.10 m child can stand over the frame, and gives lower mass. 24 in rolls better on ruts and suits riders above about 1.35 m, but raises standover to about 510 mm and fails R1 for the smallest riders. Neither is a regional roadster size, so tires and rims are the one commonality gap either way. Recommendation: 20 in. Proposed, awaiting Amish.
- **Brakes: coaster brake plus front rim brake.** A coaster brake is simple, weather-proof, needs no hand strength, and uses hub internals common on regional roadsters. On its own it fails completely if the chain comes off and is marginal on loose gravel (about 0.19 g). Adding a front rim brake gives two independent brakes. Alternatives: coaster only (cheaper, one failure mode away from no brake) or two hand brakes without a coaster (needs hand strength small children may lack). Recommendation: coaster plus front rim brake. Proposed, awaiting Amish.
- **Step-through frame.** Lower standover than a diamond frame, and practical for school uniforms and skirts. It needs twin down tubes to keep the frame stiff. Proposed, awaiting Amish.
- **Puncture-proof tires.** Solid or airless tires remove thorn punctures entirely but add about 0.8 kg, ride harsher and roll less freely. Thorn-resistant tubes with tire liners keep standard parts and a softer ride, but still puncture occasionally. Recommendation: solid or airless for the prototype, and ask users. Proposed, awaiting Amish.
- **Parts commonality.** Chain, coaster hub internals, 1 in headset, 25.4 mm seat post, 22.2 mm handlebar and 9/16 in pedal threads are chosen to match adult roadster parts. Children's bikes often use 1/2 in pedal threads; GrowRider does not.
- **Bolted clamps, not quick releases.** 13 mm bolts discourage casual changes and theft of the seat post, and every mechanic already has the spanner.

■ Safety

Safety: GrowRider is ridden by children on roads shared with trucks and motorbikes. It is a concept. No part of it has been built or tested, and it must not be ridden until the frame, fork, sliding joints, rack and brakes have passed the tests of the applicable standard (ISO 8098 or ISO 4210) at a later TRL.

- **Brakes.** Two independent brakes are required (R7). The coaster brake stops working if the chain comes off, so the front brake is not optional. Small hands need a short-reach lever. Riders need to be shown that a hard front brake on loose gravel can skid the front wheel.

- **Telescoping parts.** A seat post or stem raised past its minimum insertion can break or pull out. Every sliding part carries a permanent minimum insertion mark, and a positive stop is proposed at TRL 3. Clamp slots are pinch points for fingers.
- **Rack limit.** The rack is rated 10 kg and marked. It must not carry passengers, water containers or goods above the rating: a heavy load high over the rear wheel makes a light bike hard for a child to steer and can lift the front wheel. The rack is sized for a school bag, with no footrests.
- **Road visibility.** School trips often start and end near dawn or dusk. Front white, rear red, spoke and pedal reflectors are fitted, and a bright frame color is proposed. A light (for example a hub dynamo) is suggested for a later version.
- **Helmet.** An adjustable, ventilated child helmet is proposed with each bike (BOM line 20). Hot-climate comfort will decide whether it is worn, so it needs to be discussed in co-design. Proposed, awaiting Amish.
- **Clothing and hot surfaces.** The chainguard keeps skirts and trousers out of the chain. A light saddle cover limits heating in the sun.
- **Mass.** At about 15.5 kg, the bike is about three quarters of the weight of a 20 kg six-year-old. This makes pushing, lifting and controlling a fall harder for the smallest riders, and is the reason R5 matters.

■ Open questions for TRL 3

- Reduce mass toward 13 kg: chromoly tubes, pneumatic tires with liners, lighter rack and fenders. Which are acceptable to users and mechanics?
- Sliding joints in dust and rain: will greased steel sleeves seize over years of use? Would a split collar with a wiper seal or a plastic bushing help?
- Crank length: 140 mm is long for a 1.10 m rider and short for a 1.65 m rider. Keep one length, or offer a 127 mm swap for the smallest riders? Proposed: one length for the prototype, awaiting Amish.
- Which standard applies across the fit range, ISO 8098, ISO 4210 or both?
- Confirm with a partner which roadster parts are actually stocked in the target region, including coaster hubs and 20 in tires.
- Test the anthropometric assumptions (inseam ratio, reach) with measurements of children in the target region, where average heights may be below global references.

Concept media: [blueprint sheet](#), [interactive 3D model](#).